

SESSION 6 — REPRODUCIBILITY AUDIT

Critical Evaluation of a Machine Learning Pipeline for Digital Twins

Final English Version – Research Methods Project

1. Introduction: A Researcher's Reflection

When I first started building my Machine Learning pipeline, I believed that I already understood what reproducibility meant. I had incorporated Git for version control, DVC for dataset tracking, Docker for standardizing the execution environment, and fixed random seeds to reduce experimental variability. At that stage, I assumed that having the right tools would solve most reproducibility problems.

This audit changed that assumption. I realized that reproducibility is not simply a matter of installing the appropriate tools. It is a research discipline that requires decisions, assumptions, data transformations, software versions, computational resources, and limitations to be made explicit. Even when code and data are available, independent researchers may still struggle to reproduce a result if important methodological or environmental details remain implicit.

For this reason, I approached the audit as an exercise in self-criticism rather than as an attempt to prove that the pipeline is perfect. The purpose is to identify what an independent researcher can reproduce, what still requires additional information, and which claims should remain provisional until stronger empirical evidence is available.

2. Audit Objective

The main objective of this audit is to evaluate the reproducibility of the Machine Learning pipeline developed for a digital twin model intended to support cost and schedule optimization in construction projects. The assessment covers the source code, datasets, assumptions, computational environment, experimental workflow, and the ability of an independent researcher to reconstruct the reported process.

An important distinction is maintained throughout the audit: technical reproducibility does not automatically mean that the scientific claims have been fully validated. A pipeline may run successfully and still require additional empirical testing before its performance can be interpreted as reliable evidence.

3. Audit Methodology

The evaluation was structured around three main references identified in the project materials. First, the NeurIPS Reproducibility Checklist was used as a framework for checking whether methods, data, experiments, and results are sufficiently described. Second, the Royal Society's work on reproducibility was considered for its emphasis on practical access to code and data and the ability to repeat analyses. Third, the ML Reproducibility Auditor was considered as a reference for software-engineering quality and reproducibility signals.

To make the assessment easier to interpret, each criterion was assigned one of three statuses:

Status	Meaning
GREEN	The criterion is fully addressed and supported by sufficient documentation or evidence.

AMBER	The criterion is partially addressed, but relevant details or evidence are still missing.
RED	The criterion is not adequately addressed or documented.

4. Audit Results

4.1 Code and Algorithm Documentation

Criterion	Status	Evidence and Critical Reflection
Clear algorithm description	GREEN	The digital twin architecture is documented in the research protocol.
Complexity analysis	AMBER	Temporal and spatial complexity of the main pipeline stages has not yet been explicitly documented.
Downloadable source code	GREEN	The source code is version-controlled in a GitHub repository.
Specified dependencies	GREEN	A pinned requirements.txt file and a Dockerfile are available to standardize the software environment.

Critical reflection: The availability of source code and clearly defined dependencies is a significant strength. However, the absence of a formal complexity analysis limits the assessment of scalability. A future version should record the approximate computational cost, memory requirements, and behavior of the principal pipeline stages as the dataset grows.

4.2 Theoretical Claims and Expected Results

Criterion	Status	Evidence and Critical Reflection
Statement of expected results	GREEN	The project defines quantitative targets: MAPE below 12% and cost/schedule reductions greater than 30%.
Explicit assumptions	AMBER	Several assumptions have been identified, but they should be consolidated in a dedicated assumptions section.
Complete test of the claim	RED	Empirical validation using real projects is still pending.

Critical reflection: A performance target should not be presented as an achieved result. The MAPE threshold and the expected reduction in costs and schedules are success criteria, not evidence. They can only be

treated as demonstrated results after reproducible experiments using independent test data have been completed and the uncertainty around the estimates has been reported.

4.3 Data Collection and Processing

Criterion	Status	Evidence and Critical Reflection
Data collection description	GREEN	Technical records, project valuations, and final settlements are identified as the main data sources.
Downloadable/access-controlled data	AMBER	The datasets contain confidential information; DVC provides version tracking through a controlled storage location.
Excluded data	RED	There is not yet a formal record of excluded records or projects and the reasons for exclusion.
Preprocessing	GREEN	Imputation, standardization, and feature engineering procedures are documented.
Train/validation/test split	GREEN	A 70/15/15 split is used with <code>random_state=42</code> .

Critical reflection: The main weakness in this section is exclusion traceability. To reduce the risk of selection bias, every excluded record or project should be associated with a verifiable reason, such as missing information, inconsistent values, unavailable critical variables, or failure to meet a predefined quality criterion.

4.4 Computational Reproducibility

Criterion	Status	Evidence and Critical Reflection
Fixed random seeds	GREEN	<code>random_state=42</code> is used and the main sources of randomness are controlled.
Deterministic data split	GREEN	The dataset partition uses a fixed random seed.
Documented software environment	GREEN	The Dockerfile and <code>requirements.txt</code> support reconstruction of the software environment.
Documented infrastructure	RED	The hardware environment has not yet been completely recorded, including CPU, GPU, RAM, storage, and driver versions.

Critical reflection: Fixed seeds are necessary, but they do not guarantee perfect reproducibility. Differences may still arise from hardware, numerical libraries, parallel execution, or driver versions. Therefore, the computational environment should document both software and infrastructure.

5. Overall Audit Score

Category	GREEN	AMBER	RED
Code and algorithm	3	1	0
Theoretical claims	1	1	1
Data	3	1	1
Reproducibility	3	0	1
TOTAL	10	3	3

Interpretation: The audit indicates a solid reproducibility foundation, particularly in source-code management, dependency definition, data processing, and control of random seeds. At the same time, six criteria are not yet green. These represent concrete risks: the lack of a complexity analysis, incomplete documentation of assumptions, pending real-world validation, insufficient traceability of excluded data, and incomplete hardware documentation.

6. The Stranger Test

The most demanding practical test is to imagine that the repository is handed to a researcher who did not participate in its development. If that person can clone the repository, obtain the authorized datasets, rebuild the environment, execute the pipeline, and obtain consistent outputs without contacting the original researcher, there is strong practical evidence of reproducibility.

Step	Can a stranger do it?	Evidence
Clone the repository	Yes	GitHub repository with documented execution instructions.
Obtain the data	Yes, with restrictions	DVC can recover datasets from the configured secure storage location.
Build the environment	Yes	A Dockerfile is available and has been tested.
Execute the pipeline	Yes	The pipeline can be executed using the documented command.
Reproduce the results	Partially	Technical determinism is supported, but real-project validation is not yet complete.

The result of the stranger test is therefore encouraging but not conclusive. The workflow is technically accessible, yet the scientific claim remains partly dependent on validation that has not been completed.

7. Main Reproducibility Gaps

- Document the complete computational infrastructure, including CPU, GPU, RAM, storage, operating system, drivers, and relevant software versions.
- Create a formal assumptions table describing each assumption, its justification, and its possible effect on the results.
- Maintain a complete record of excluded records and projects, together with objective exclusion criteria.
- Complete empirical validation using real projects while preserving a separate final test set.
- Report confidence intervals, standard deviations, or other suitable uncertainty measures for the main metrics.
- Perform temporal and spatial complexity analysis for the principal pipeline stages.
- Record Python, CUDA/cuDNN where applicable, numerical library, operating system, and driver versions.
- Introduce automated reproducibility tests to detect unexpected changes in data, models, metrics, and generated outputs.

8. Reproducibility Improvement Plan

Priority	Action	Expected Outcome
High	Document hardware and software completely.	More faithful reconstruction of the experimental environment.
High	Complete validation with real projects.	Determine whether the target metrics remain achievable outside the development dataset.
High	Calculate confidence intervals.	Quantify uncertainty and improve interpretation of model performance.
Medium	Document exclusions and assumptions.	Reduce ambiguity and potential sources of selection or methodological bias.
Medium	Analyze computational complexity.	Assess scalability and execution cost.
Medium	Automate reproducibility tests.	Detect regressions and unintended changes in the pipeline.

9. Conclusion

This audit shows that the pipeline has a reasonably strong reproducibility structure. Version control, DVC-based data management, Docker-based environment definition, fixed random seeds, and documented preprocessing provide a useful foundation for independent reconstruction.

However, reproducibility should not be treated as a simple yes-or-no property. A project can be technically executable while still containing methodological uncertainties that prevent its results from being interpreted

as definitive evidence. In this project, the most important limitations are the incomplete validation with real projects, the absence of uncertainty intervals, incomplete hardware documentation, and insufficient traceability of excluded data.

The most important lesson from this session is that reproducibility means making explicit the details that researchers often leave implicit. An independent researcher should not need to contact the original author to discover which library version was used, which observations were excluded, which hardware executed the model, or which assumptions shaped the analysis. The next stage of the project should therefore turn the findings of this audit into documented and verifiable controls.

This process has also changed my view of what it means for a Machine Learning project to be research-ready. A model is not sufficiently documented simply because it produces a good metric. Its data, assumptions, environment, evaluation procedure, and limitations must be visible enough for another person to question, reproduce, and improve the work. That is the standard I intend to apply to the next stage of this digital twin research.

10. References

1. Pineau, J., et al. (2021). Improving reproducibility in machine learning research: A report from the NeurIPS 2019 Reproducibility Program. *Journal of Machine Learning Research*, 22(164), 1–20.
2. Royal Society. (2026). Code sharing and reproducibility in survey-based social research: Evidence from a large-scale audit. Royal Society Open Science.
3. ML Reproducibility Auditor. (2026). ML Systems Reproducibility Auditor documentation. PyPI.

11. AI Use Disclosure Statement

AI-assisted tools were used during the preparation of this document for language refinement, organization, and clarification of the material. The underlying audit criteria, findings, status classifications, scores, limitations, and project-specific observations were preserved from the source material. The final document was reviewed and adapted by the researcher, who remains responsible for the accuracy, interpretation, and academic integrity of the submitted work.